

# Marking Scheme

2026 · Paper 1 · 105 marks

## General Rules

- M / A / PROOF mark types
- **f.t.**：承上題錯誤但方法正確者，可獲後續步驟分。
- Deduction cap：A(1)=1 · A(2)=1 · B=0 (u, pp)
- u = 單位遺漏或錯誤；pp = 準確度不符合要求。每部分最多扣 1 分。
- 答案正確且步驟合理者可獲滿分。
- 符號書寫略有出入但意思清楚者不扣分。 · Benefit of doubt
- 分數、小數或根式形式均接受，除非題目另有指明。

## Section A(1) (35 marks)

### 1. Change of Subject [B-]

a (3 marks)

**M1**  $6b - 15c = 4b + 9$

**M1**  $2b = 15c + 9$  **f.t.**

**A1**  $b = \frac{15c + 9}{2}$  **f.t.**

### 2. Laws of Indices [B]

a (3 marks)

**M1**  $= \frac{m^6 n^{-4}}{m^{-1} n^4}$

**M1**  $= m^{6-(-1)} n^{-4-4} = m^7 n^{-8}$  **f.t.**

**A1**  $= \frac{m^7}{n^8}$  **f.t.**

### 3. Approximation [B]

a (1 mark)

**A1** **373**

b (1 mark)

**A1** **372.6**

c (1 mark)

**A1** **370**

### 4. Probability [B]

a (3 marks)

**M1**  $\frac{9}{n+6+9} = \frac{3}{8}$

**M1**  $3(n+15) = 72$  **f.t.**

**A1**  $n = 9$  **f.t.**

## 5. Factorization [B]

a (2 marks)

M1  $difference of squares = (2r)^2 - (3s)^2$

A1  $(2r - 3s)(2r + 3s)$

b (2 marks)

M1  $= (2r - 3s)(2r + 3s) - (2r + 3s)$  f.t.

A1  $= (2r + 3s)(2r - 3s - 1)$  f.t.

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## 6. Inequalities [B+]

a (3 marks)

M1  $3x - 1 < 2x + 8 \Rightarrow x < 9$

M1  $x \geq -7$

A1  $-7 \leq x < 9$  f.t.

b (1 mark)

A1  $8$  f.t.

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## 7. Percentages [B+]

a (5 marks)

M1  $Let cost be \textcolor{red}{C}; marked price = 1.3C$

M1  $selling price = 1.3C \times (1 - 40\%) = 0.78C$  f.t.

M1  $C - 0.78C = 88$  f.t.

A1  $cost = \textcolor{red}{400}$  f.t.

A1  $marked price = \textcolor{red}{520}$  f.t.  
*u-1 if \$ sign missing*

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## 8. Properties of Circles [B+]

a (5 marks)

M1  $AB \parallel ED \Rightarrow \widehat{AE} = \widehat{BD}$

M1  $x = \angle ABE = \angle BAD$  (equal arcs, equal inscribed angles) f.t.

A1  $x = \theta$  f.t.

M1  $In \triangle ABF: y = 180^\circ - \theta - x$  f.t.

A1  $y = 180^\circ - 2\theta$  f.t.

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## 9. Speed (Applications) [B+]

a (5 marks)

M1  $Let PQ = d km, QR = (210 - d) km$

M1  $\frac{d}{72} + \frac{210 - d}{90} = \frac{161}{60}$  f.t.

M1  $Solving, d = 126$  f.t.

M1  $time = \frac{126}{72} h$  f.t.

A1  $= 1\frac{3}{4} h = 105 minutes$  f.t.  
*u-1 if unit missing*

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## Section A(2) (35 marks)

### 10. Box-and-Whisker Diagrams [A-]

a (3 marks)

M1  $Range : b - 15 = 40$

A1  $b = 55$  f.t.

A1  $IQR : a - 23 = 19 \Rightarrow a = 42$  f.t.

b (2 marks)

M1 *TheagerangeofYisnotfixed(e.g. 38to58), and58 > 55*

A1 *Disagree : thecombinedrangemayexceed40* f.t.

### 11. Statistical Measures [A-]

a (2 marks)

A1 (i)  $least k = 1$   
需10 為唯一眾數 / 10 must be the unique mode

A1 (ii)  $greatest k = 9$

b (2 marks)

A1 (i)  $least k = 1$

A1 (ii)  $greatest k = 14$

c (2 marks)

M1  $\frac{k + 44}{21 + k} = 2$

A1  $k = 2$  f.t.

### 12. Remainder Theorem [A]

a (3 marks)

M1  $f(2) = 0$  and  $f(-1) = 15$

M1  $2a + b + 4 = 0, -a + b - 8 = 15$  f.t.

A1  $a = -9, b = 14$  f.t.

b (4 marks)

M1  $f(x) = (x - 2)(2x^2 - 7)$  f.t.

M1  $2x^2 - 7 = 0 \Rightarrow x = \pm\sqrt{\frac{7}{2}}$  f.t.

A1  $\sqrt{7/2}$  is irrational f.t.

A1 Agree (irrational root exist) f.t.

### 13. Similar Triangles [A]

a (2 marks)

**PROOF**  $\angle ABE = \angle ECD = 90^\circ; \angle AEB = 90^\circ - \angle DEC = \angle EDC; \Rightarrow \triangle ABE \sim \triangle ECD(AAA).$

**A2** *correctproof*

b (6 marks)

**M1**  $BE = \sqrt{20^2 - 12^2} = 16; \frac{AB}{EC} = \frac{BE}{CD}$

**A1** (i)  $CD = 40cm$  f.t.

**M1**  $ED = \frac{20 \times 30}{12} = 50$  f.t.

**A1** (ii)  $area = 500cm^2$  f.t.

**M1**  $shortestdistance = \frac{2(500)}{10\sqrt{29}} \approx 18.6$  f.t.

**A1** (iii)  $18.6 < 23, so yessuch F exists$  f.t.

### 14. Volume of Solids [A]

a (2 marks)

**M1**  $V = \pi(8)^2(64)$

**A1**  $= 4096\pi cm^3$  f.t.

b (4 marks)

**M1**  $watersurfacradiusr = \frac{h}{3}$

**M1**  $\frac{1}{3}\pi\left(\frac{h}{3}\right)^2 h = 4096\pi$  f.t.

**M1**  $h^3 = 110592$  f.t.

**A1**  $h = 48cm$  f.t.  
*u-1 if unit missing*

c (3 marks)

**M1**  $sphere = \frac{4}{3}\pi(14)^3 = \frac{10976}{3}\pi$

**M1**  $water + sphere = \frac{23264}{3}\pi; vesselcapacity = 8000\pi$  f.t.

**A1**  $\frac{23264}{3}\pi < 8000\pi, so the water will not overflow$  f.t.

## Section B (35 marks)

### 15. Permutations [A]

a (1 mark)

**A1**  $8! = 40320$

b (2 marks)

**M1** *odddigits* 3, 5, 7, 9 : *ends*  $4 \times 3$ , *middle* 6!

**A1**  $4 \times 3 \times 6! = 8640$  f.t.

## 16. Geometric Sequences [A]

a (2 marks)

M1  $r = 3; 1st term = \frac{45}{3^2}$

A1  $= 5$  f.t.

b (3 marks)

M1  $5 \cdot 3^n + 5 \cdot 3^{2n} < 5 \times 10^{14}$  f.t.

M1  $3^n + 3^{2n} < 10^{14}; n = 14 \text{ works}, n = 15 \text{ fails}$  f.t.

A1  $greatest n = 14$  f.t.

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## 17. Trigonometry in 3-D [A+]

a (2 marks)

M1  $\angle ADB = 40^\circ; \frac{AD}{\sin 20^\circ} = \frac{60}{\sin 40^\circ}$

A1  $AD \approx 31.9cm$  f.t.  
r.t. 31.9

b (5 marks)

M1  $\cos \angle ABC = \frac{60^2 + 31.9^2 - 40^2}{2(60)(31.9)}$  f.t.

A1 (i)  $\angle ABC \approx 38.0^\circ$  f.t.  
r.t. 38.0

M1 *perpendicular distance h from A, C to BD* f.t.

M1  $\cos \phi = \frac{h^2 + h^2 - 40^2}{2h^2}$  f.t.

A1 (ii) *angle between planes*  $\approx 71.9^\circ$  f.t.  
r.t. 71.9

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## 18. Partial Variation [A+]

a (3 marks)

M1 *Let*  $f(x) = ax^2 + bx : a + b = -5, 49a + 7b = 7$

M1  $a = 1, b = -6$  f.t.

A1  $f(x) = x^2 - 6x$  f.t.

b (5 marks)

M1 (i)  $f(x) = (x - 3)^2 - 9$  f.t.

A1 (i)  $Q = (3, -9)$  f.t.

A1 (ii)  $R = (3, 36)$  f.t.

M1 (iii)  $\vec{SQ} \cdot \vec{SR} = 0$ , so  $\angle QSR = 90^\circ$

A1 (iii) *Q is a diameter of the circumcircle, so P is the mid-point of QR* f.t.

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## 19. Equations of Circles [A+]

**a** (4 marks)

**A1**  $C : (x - 8)^2 + (y - 2)^2 = r^2$

**M1**  $tangent \Rightarrow \frac{|8k - 10 - 21|}{\sqrt{k^2 + 25}} = r$

**M1**  $\frac{|8k - 31|}{\sqrt{k^2 + 25}} = r$  **f.t.**

**A1**  $r^2 = \frac{(8k - 31)^2}{k^2 + 25}$  **f.t.**

**b** (8 marks)

**M1** (i) *Put D(18, 39) into L* :  $18k - 195 - 21 = 0$

**A1** (i)  $k = 12$  **f.t.**

**M1** (i)  $r^2 = \frac{4225}{169} = 25$  **f.t.**

**A1** (i)  $r = 5$  **f.t.**  
*u-1 if unit issues*

**M1** (ii)  $E = (0, -\frac{21}{5})$  **f.t.**

**M1** (ii) *other tangents from D, E meet at F*  $\approx (12.5, -2.4)$  **f.t.**

**M1** (ii) *largest angle*  $\angle DFE \approx 105.8^\circ$  **f.t.**

**A1** (ii)  $105.8^\circ > 90^\circ$ , *so  $\triangle DEF$  is obtuse – angled* (yes) **f.t.**

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